

Gas Monitoring System using Arduino and LCD

Arduino Program

```
#include <LiquidCrystal.h>

LiquidCrystal lcd(8, 7, 6, 5, 4, 3);

const int gasPin    = A0;
const int buzzerPin = 9;
const int motorPin  = 10;
const int ledPin    = 11;

int gasValue = 0;

void setup() {
  pinMode(buzzerPin, OUTPUT);
  pinMode(motorPin, OUTPUT);
  pinMode(ledPin, OUTPUT);

  Serial.begin(9600);
  lcd.begin(16, 2);

  lcd.setCursor(0, 0);
  lcd.print(" GAS MONITOR ");
}

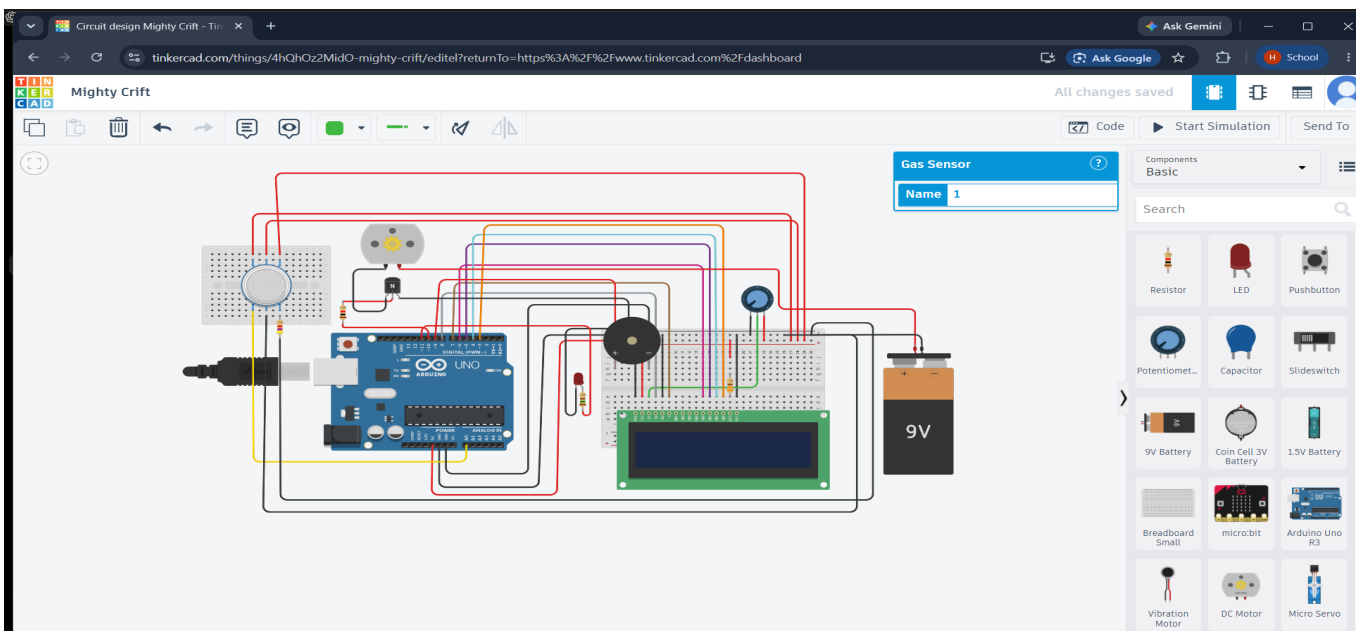
void loop() {
  gasValue = analogRead(gasPin);

  Serial.print("Gas Value: ");
  Serial.println(gasValue);

  lcd.setCursor(0, 1);
  lcd.print(" GAS MONITER      ");
  lcd.setCursor(0, 1);

  if (gasValue > 300) {
    lcd.print("Gas Detected");
    digitalWrite(buzzerPin, HIGH);
    digitalWrite(motorPin, HIGH);
    digitalWrite(ledPin, HIGH);
  }
  else {
    lcd.print("No Gas Detected");
    digitalWrite(buzzerPin, LOW);
    digitalWrite(motorPin, LOW);
    digitalWrite(ledPin, LOW);
  }

  delay(200);
}
```



Explanation:

Project Objective:

This Arduino-based Gas Monitoring System continuously monitors the gas concentration using a gas sensor. When the gas level exceeds the preset threshold value (300), the system alerts the user through a buzzer, LED indicator, and motor (exhaust fan). The current status is displayed on a 16x2 LCD.

Code Explanation:

1. The LiquidCrystal library is included to interface the 16x2 LCD display.
2. LCD pins are initialized using the LiquidCrystal object.
3. Gas sensor is connected to analog pin A0.
4. Buzzer, motor, and LED are connected to digital pins 9, 10, and 11 respectively.
5. In setup(), the output devices are configured and the LCD is initialized.
6. The message "GAS MONITOR" is displayed on the LCD during startup.
7. In loop(), the gas sensor value is read using analogRead().
8. The sensor reading is printed on the Serial Monitor for debugging and monitoring.
9. The LCD second line is cleared before displaying the current status.
10. If the gas value is greater than 300, the system displays "Gas Detected".
11. During gas detection, the buzzer, LED, and motor are turned ON to provide warning and ventilation.
12. If the gas value is below the threshold, "No Gas Detected" is displayed and all outputs remain OFF.
13. A delay of 200 ms is provided before the next reading.

Working Principle:

- The gas sensor continuously senses the surrounding air.
- Arduino converts the analog sensor output into a digital value.
- The value is compared with the threshold value (300).
- If gas leakage is detected, alarm devices are activated automatically.
- The LCD provides real-time status information to the user.

Applications:

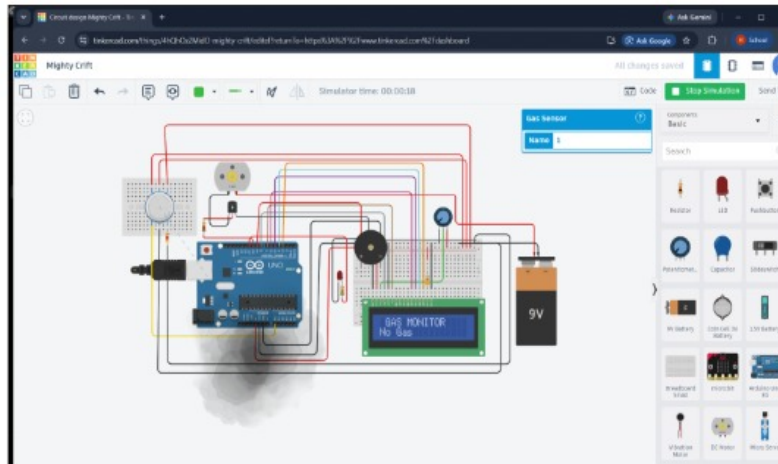
- Home gas leakage detection
- Industrial safety monitoring
- LPG storage areas
- Chemical laboratories

Tinkercad link:

<https://www.tinkercad.com/things/4hQhOz2MidO-mighty-crift>

Output:

When no gas detected:



When gas detected:

